# Alex Doe

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Springfield, ST | Willing to relocate

## Education

**B.S. Electrical Engineering**, GPA 3.62/4.00 Expected Spring 2027
Example State University
**Relevant coursework:** Embedded Systems, Introduction to High Power Electronics, Advanced Digital Design
**Activities:** IEEE student branch, Tau Beta Pi

## Projects

**Synchronous Buck Converter for a CubeSat Power Board** Fall 2025–Spring 2026
KiCad, LTspice, STM32, C

- Raised peak efficiency from 85% to 92% at 2 A on an electronic load by replacing the catch diode with a synchronous MOSFET and tuning dead time in 10 ns steps on an STM32 timer
- Brought output ripple down from 80 mV to 25 mV peak to peak on the oscilloscope by moving to a 4-layer KiCad layout with a solid ground plane and a smaller switching loop, simulated in LTspice
- Wrote overcurrent shutdown firmware in C that trips in under 50 µs, tested on 20 shorts

**FPGA Audio Equalizer** Spring 2025
Verilog, Vivado, MATLAB, Basys 3

- Built a 5-band equalizer in Verilog for 48 kHz stereo audio with under 2 ms of latency
- Kept the response within 0.5 dB of the MATLAB reference from 20 Hz to 20 kHz by testing 16-bit fixed-point coefficients in Vivado simulation before loading the design onto a Basys 3 board

**Bluetooth Battery Monitor** Fall 2024
ESP32, C++, Python, INA219

- Built an ESP32 monitor that streams 4 cell voltages to a Python dashboard over Bluetooth
- Calibrated an INA219 current sensor to within 1% of a bench meter from 0.1 A to 3 A
- Stretched field-test runtime 18% with a C++ cutoff that drops extra loads at 3.3 V per cell

## Experience

**Power Subsystem Lead, Example State CubeSat Team**, Springfield, ST Sep 2024–Present

- Saved 40 g on the power board by swapping 3 linear regulators for 1 buck converter
- Caught 2 battery protection faults before launch review with a Python script that ran overnight charge and discharge cycles on a bench supply and flagged any cell that drifted more than 20 mV
- Run weekly design reviews for the 4-person power team

Sales Associate, Example Hardware Store May 2023–Aug 2024

## Skills

**Software:** KiCad, LTspice, MATLAB, Vivado, Python, C, C++, Verilog
**Hardware/Lab:** Oscilloscope, Electronic load, Bench supply, STM32, ESP32, FPGA
**Certifications:** IPC J-STD-001 Soldering